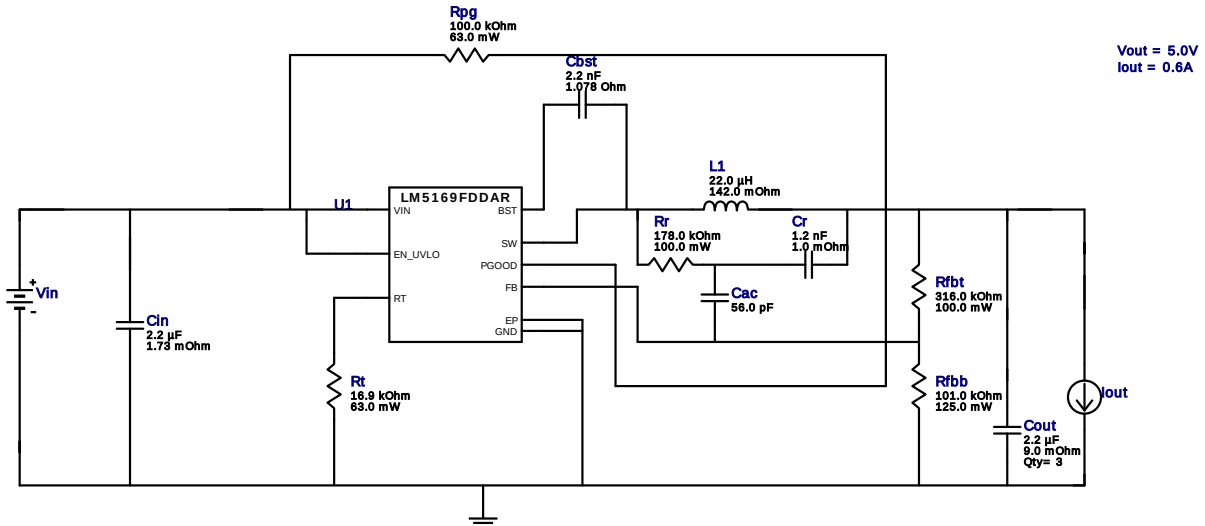


WEBENCH® Design Report

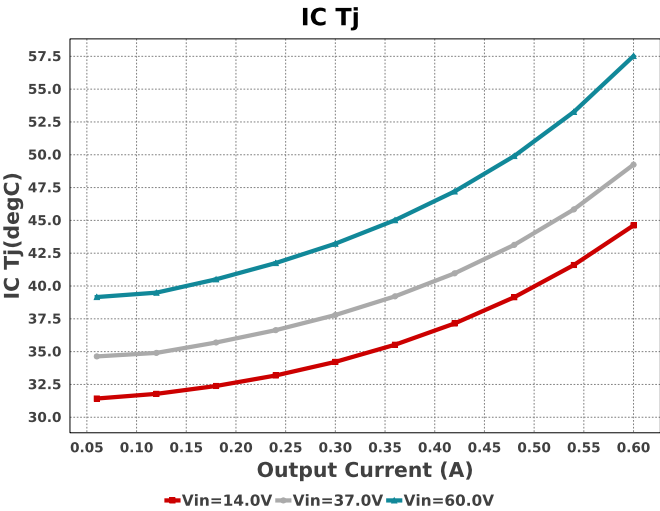
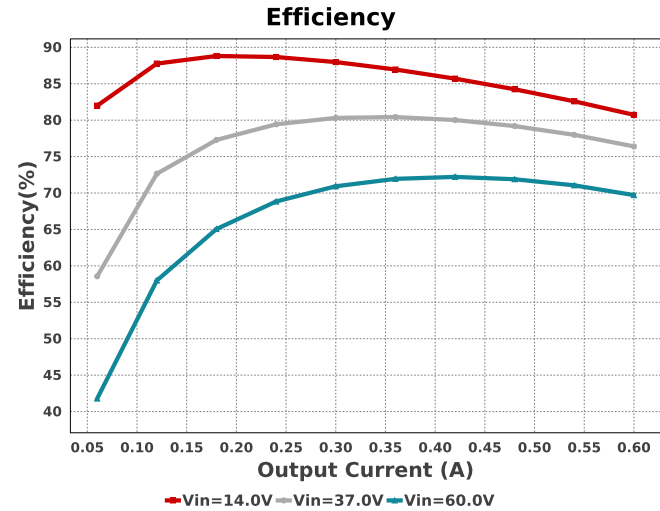
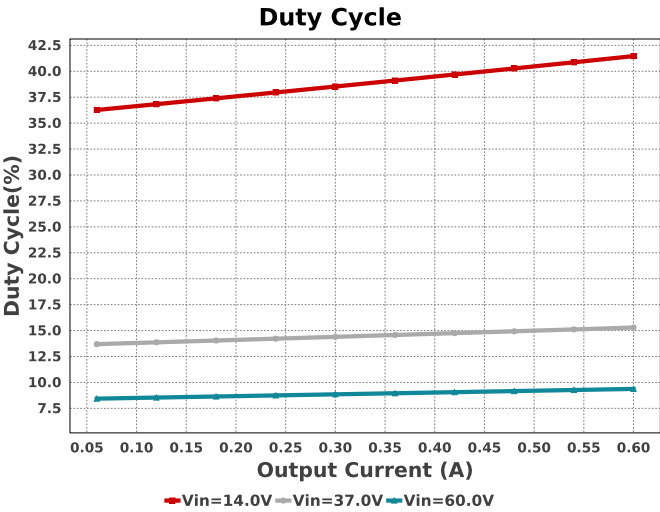
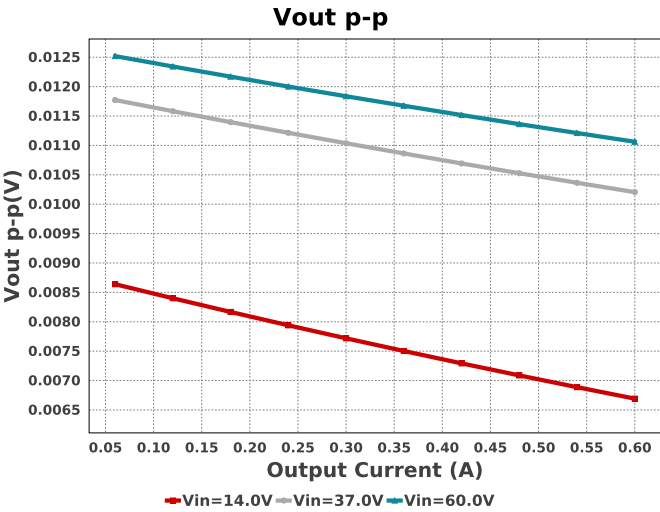
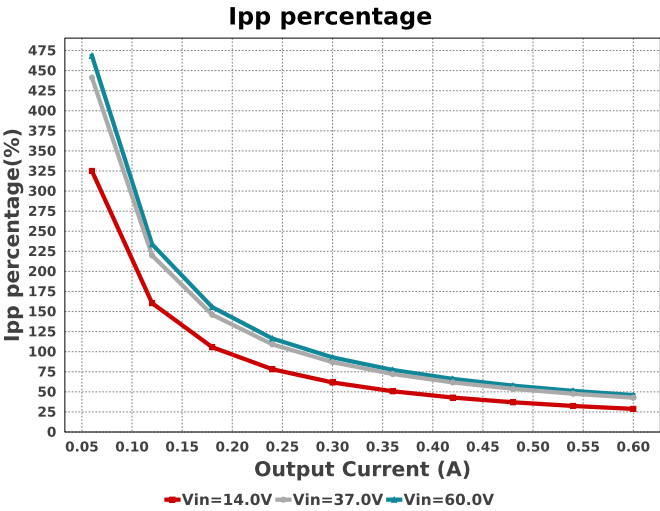
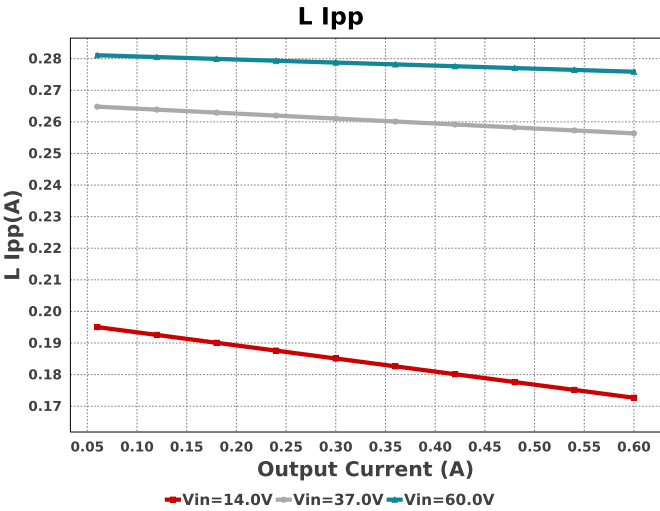
Design : 5 LM5169FDDAR
LM5169FDDAR 14V-60V to 5.00V @ 0.6A

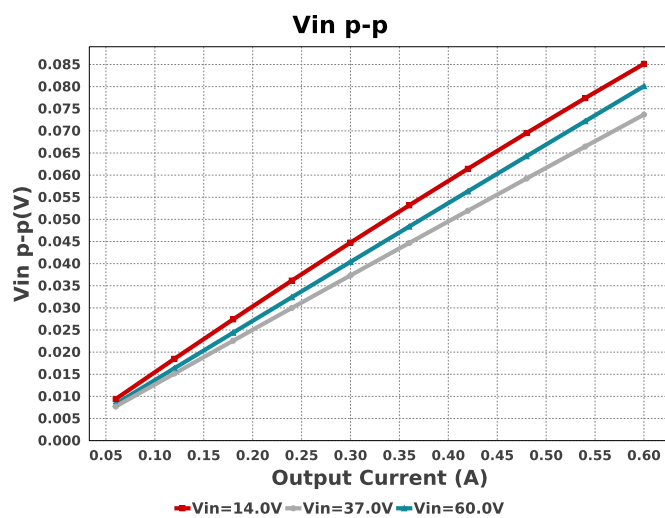
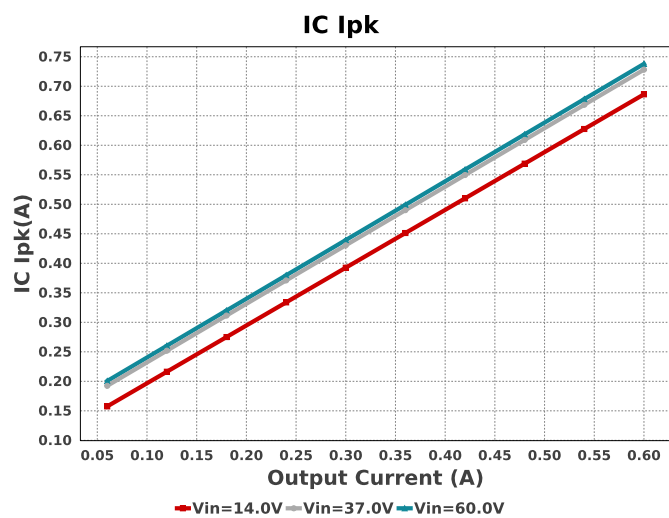
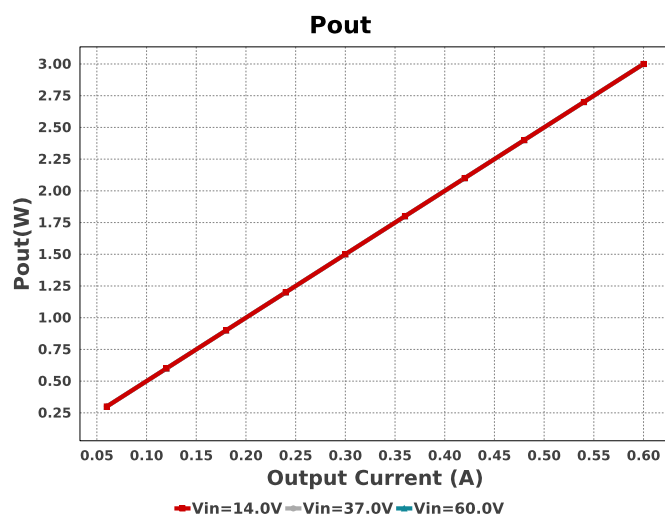
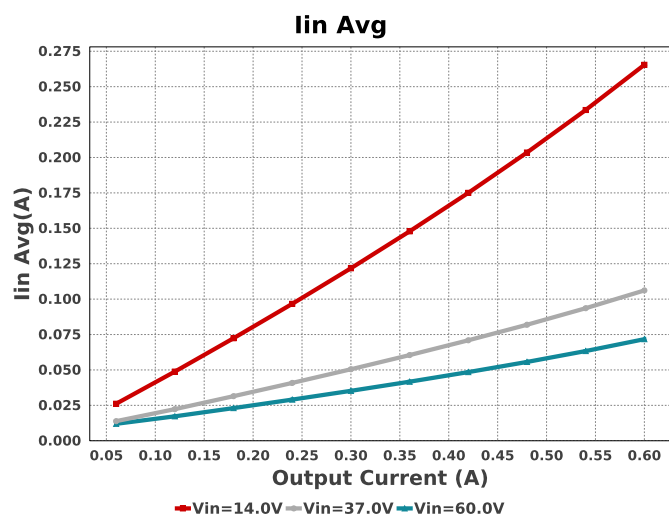
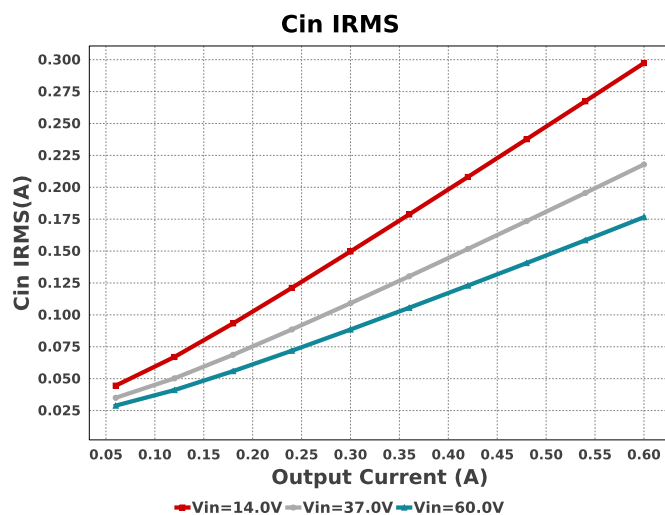
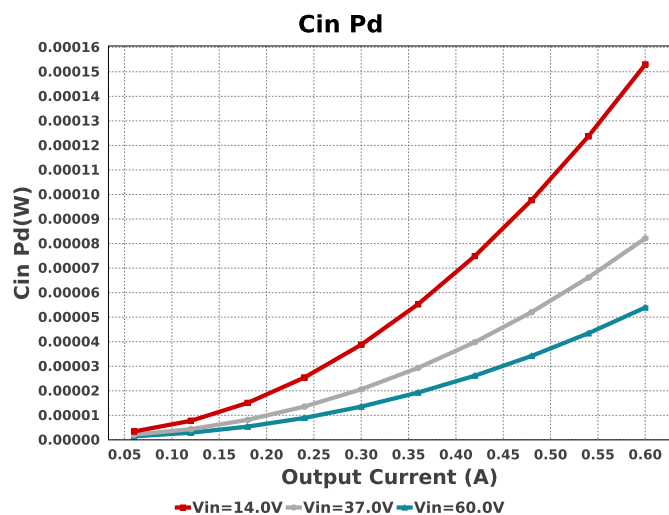


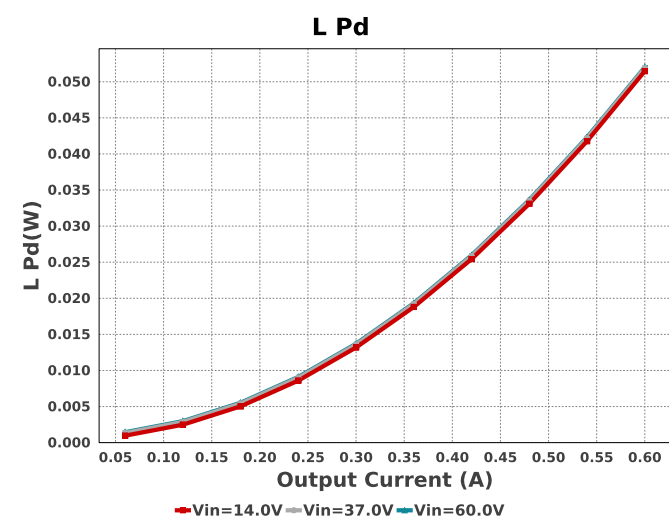
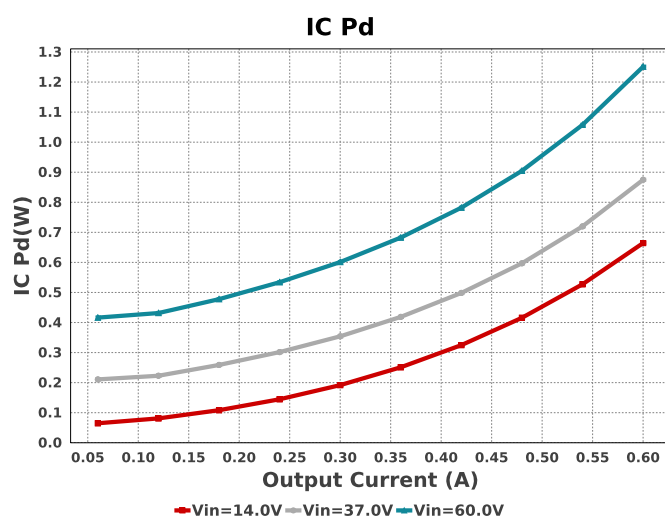
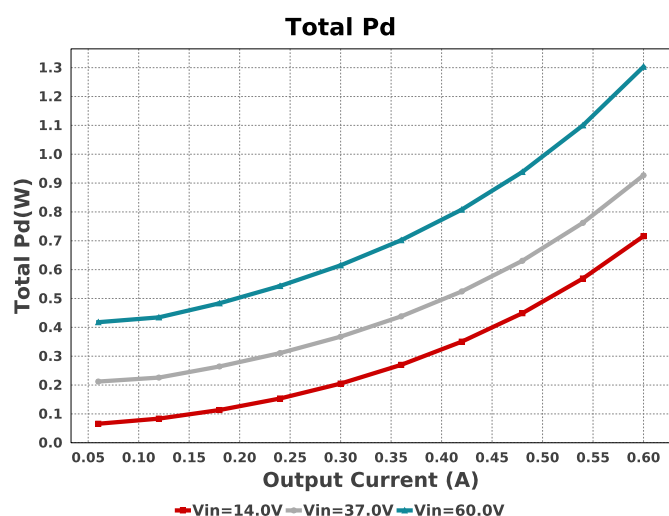
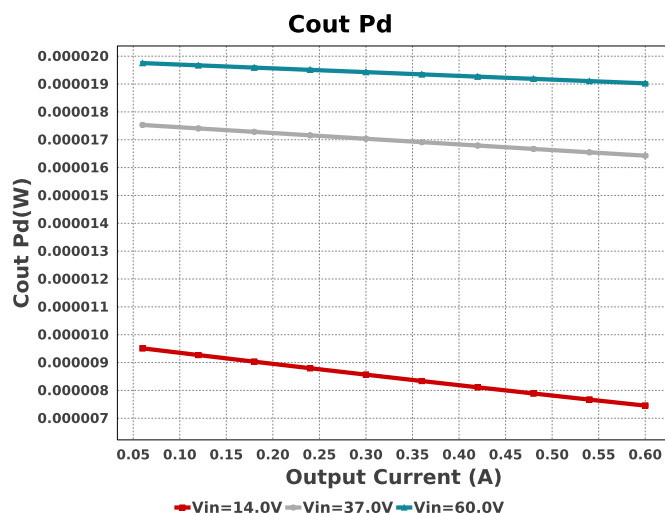
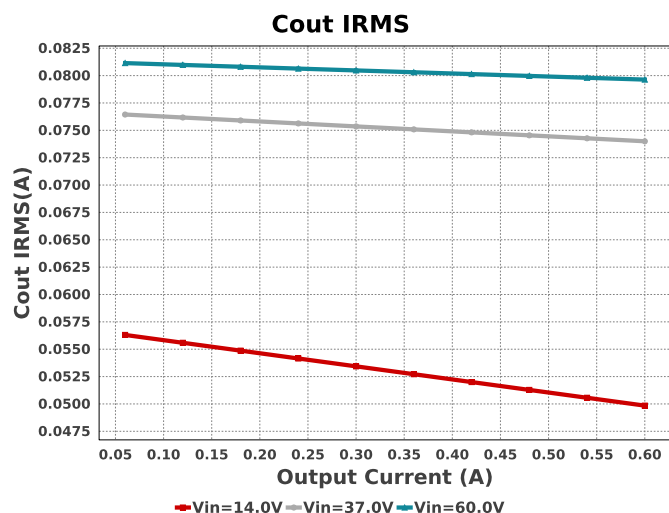
Electrical BOM

Name	Manufacturer	Part Number	Properties	Qty	Price	Footprint
Cac	TDK	CGA1A2C0G1H560J030BA Series= C0G/NP0	Cap= 56.0 pF VDC= 50.0 V IRMS= 0.0 A	1	\$0.01	0201 2 mm ²
Cbst	TDK	CGA2B2X7R1H222K050BA Series= X7R	Cap= 2.2 nF ESR= 1.0779 Ohm VDC= 50.0 V IRMS= 264.071 mA	1	\$0.01	0402 3 mm ²
Cin	TDK	C3225X7R2A225K230AB Series= X7R	Cap= 2.2 uF ESR= 1.73 mOhm VDC= 100.0 V IRMS= 5.5932 A	1	\$0.21	1210_250 15 mm ²
Cout	MuRata	GRM188R71A225KE15D Series= X7R	Cap= 2.2 uF ESR= 9.0 mOhm VDC= 10.0 V IRMS= 3.3 A	3	\$0.02	0603 5 mm ²
Cr	MuRata	GRM155R71H122KA01D Series= X7R	Cap= 1.2 nF ESR= 1.0 mOhm VDC= 50.0 V IRMS= 0.0 A	1	\$0.01	0402 3 mm ²
L1	Bourns	SRN6045-220M	L= 22.0 uH 142.0 mOhm	1	\$0.25	SRN6045 64 mm ²
Rfbb	Yageo	RT0805BRD07101KL Series= RT0805	Res= 101.0 kOhm Power= 125.0 mW Tolerance= 0.1%	1	\$0.05	0805 7 mm ²
Rfbb	Yageo	RC0603FR-07316KL Series= ?	Res= 316.0 kOhm Power= 100.0 mW Tolerance= 1.0%	1	\$0.01	0603 5 mm ²
Rpg	Vishay-Dale	CRCW0402100KFKED Series= CRCW..e3	Res= 100.0 kOhm Power= 63.0 mW Tolerance= 1.0%	1	\$0.01	0402 3 mm ²

Name	Manufacturer	Part Number	Properties	Qty	Price	Footprint
Rr	Vishay-Dale	CRCW0603178KFKEA Series= CRCW..e3	Res= 178.0 kOhm Power= 100.0 mW Tolerance= 1.0%	1	\$0.01	 0603 5 mm²
Rt	Vishay-Dale	CRCW040216K9FKED Series= CRCW..e3	Res= 16.9 kOhm Power= 63.0 mW Tolerance= 1.0%	1	\$0.01	 0402 3 mm²
U1	Texas Instruments	LM5169FDDAR	Switcher	1	\$0.96	 DDA0008B 55 mm²







Operating Values

#	Name	Value	Category	Description
1.	BOM Count	14		Total Design BOM count
2.	Total BOM	\$1.606		Total BOM Cost
3.	Cin IRMS	176.574 mA	Capacitor	Input capacitor RMS ripple current
4.	Cin Pd	53.939 μ W	Capacitor	Input capacitor power dissipation
5.	Cout IRMS	79.636 mA	Capacitor	Output capacitor RMS ripple current
6.	Cout Pd	19.025 μ W	Capacitor	Output capacitor power dissipation
7.	IC IpK	737.933 mA	IC	Peak switch current in IC
8.	IC Pd	1.251 W	IC	IC power dissipation
9.	IC Tj	57.525 degC	IC	IC junction temperature
10.	IC Tolerance	19.0 mV	IC	IC Feedback Tolerance
11.	ICThetaJA Effective	22.0 degC/W	IC	Effective IC Junction-to-Ambient Thermal Resistance

#	Name	Value	Category	Description
12.	Iin Avg	71.721 mA	IC	Average input current
13.	Ipp percentage	45.978 %	Inductor	Inductor ripple current percentage (with respect to average inductor current)
14.	L Ipp	275.866 mA	Inductor	Peak-to-peak inductor ripple current
15.	L Pd	52.021 mW	Inductor	Inductor power dissipation
16.	Cin Pd	53.939 μ W	Power	Input capacitor power dissipation
17.	Cout Pd	19.025 μ W	Power	Output capacitor power dissipation
18.	IC Pd	1.251 W	Power	IC power dissipation
19.	L Pd	52.021 mW	Power	Inductor power dissipation
20.	Total Pd	1.303 W	Power	Total Power Dissipation
21.	Duty Cycle	9.374 %	System	Duty cycle
22.	Efficiency	69.714 %	System	Steady state efficiency
23.	FootPrint	178.0 mm ²	System	Total Foot Print Area of BOM components
24.	Frequency	832.041 kHz	System	Switching frequency
25.	Iout	600.0 mA	System	Iout operating point
26.	Mode	CCM	System	Conduction Mode
27.	Pout	3.0 W	System	Total output power
28.	Vin	60.0 V	System	Vin operating point
29.	Vin p-p	80.095 mV	System	Peak-to-peak input voltage
30.	Vout	5.0 V	System	Operational Output Voltage
31.	Vout Actual	4.954 V	System	Vout Actual calculated based on selected voltage divider resistors
32.	Vout Tolerance	2.431 %	System	Vout Tolerance based on IC Tolerance (no load) and voltage divider resistors if applicable
33.	Vout p-p	11.062 mV	System	Peak-to-peak output ripple voltage

Design Inputs

Name	Value	Description
Iout	600.0 m	Maximum Output Current
VinMax	60.0	Maximum input voltage
VinMin	14.0	Minimum input voltage
Vout	5.0	Output Voltage
base_pn	LM5169F-DDA	Base Product Number
source	DC	Input Source Type
Ta	30.0	Ambient temperature

WEBENCH® Assembly

Component Testing

Some published data on components in datasheets such as Capacitor ESR and Inductor DC resistance is based on conservative values that will guarantee that the components always exceed the specification. For design purposes it is usually better to work with typical values. Since this data is not always available it is a good practice to measure the Capacitance and ESR values of C_{in} and C_{out} , and the inductance and DC resistance of $L1$ before assembly of the board. Any large discrepancies in values should be electrically simulated in WEBENCH to check for instabilities and thermally simulated in WebTHERM to make sure critical temperatures are not exceeded.

Soldering Component to Board

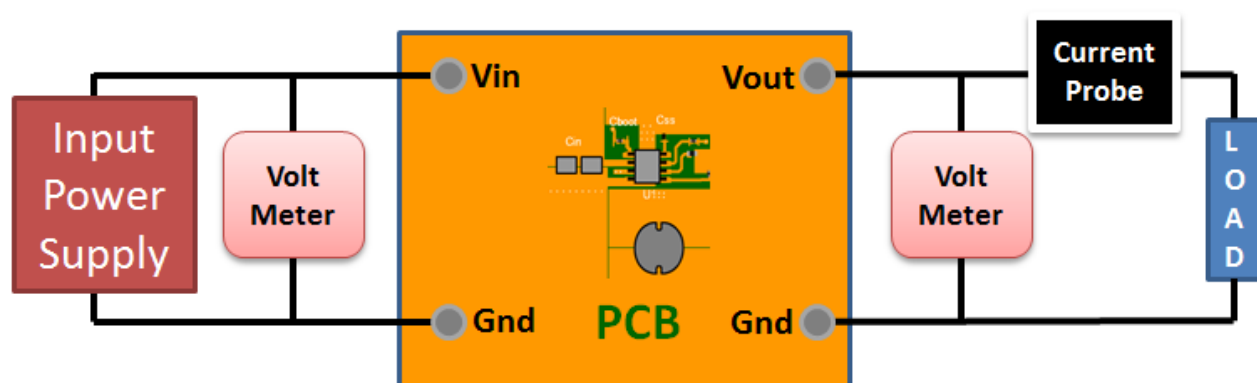
If board assembly is done in house it is best to tack down one terminal of a component on the board then solder the other terminal. For surface mount parts with large tabs, such as the DPAK, the tab on the back of the package should be pre-tinned with solder, then tacked into place by one of the pins. To solder the tab down to the board place the iron down on the board while resting against the tab, heating both surfaces simultaneously. Apply light pressure to the top of the plastic case until the solder flows around the part and the part is flush with the PCB. If the solder is not flowing around the board you may need a higher wattage iron (generally 25W to 30W is enough).

Initial Startup of Circuit

It is best to initially power up the board by setting the input supply voltage to the lowest operating input voltage 14.0V and set the input supply's current limit to zero. With the input supply off connect up the input supply to V_{in} and GND. Connect a digital volt meter and a load if needed to set the minimum load of the design from V_{out} and GND. Turn on the input supply and slowly turn up the current limit on the input supply. If the voltage starts to rise on the input supply continue increasing the input supply current limit while watching the output voltage. If the current increases on the input supply, but the voltage remains near zero, then there may be a short or a component misplaced on the board. Power down the board and visually inspect for solder bridges and recheck the diode and capacitor polarities. Once the power supply circuit is operational then more extensive testing may include full load testing, transient load and line tests to compare with simulation results.

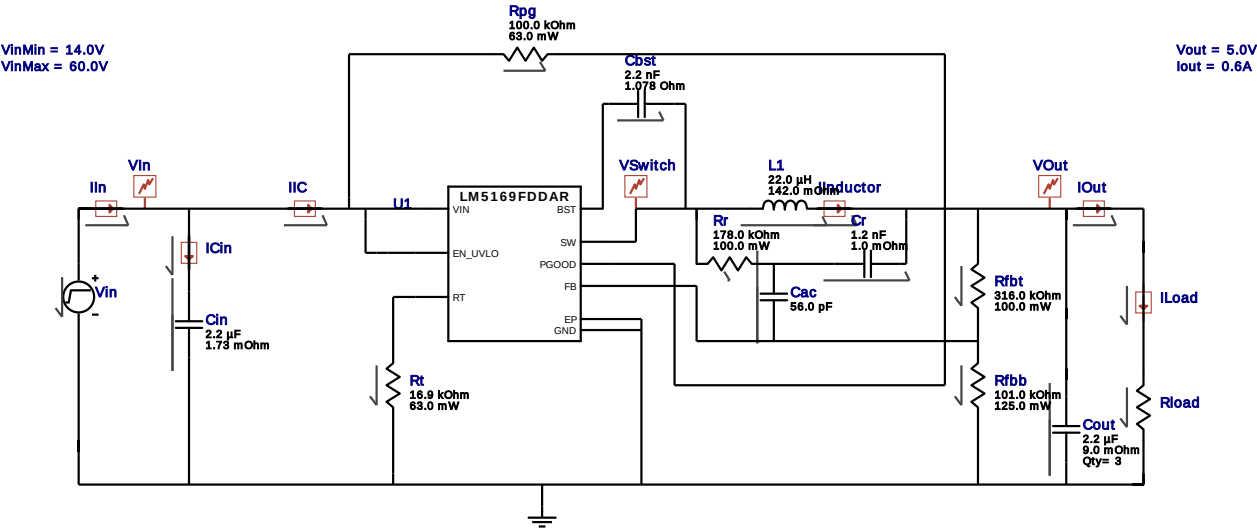
Load Testing

The setup is the same as the initial startup, except that an additional digital voltmeter is connected between V_{in} and GND, a load is connected between V_{out} and GND and a current meter is connected in series between V_{out} and the load. The load must be able to handle at least rated output power + 50% (7.5 watts for this design). Ideally the load is supplied in the form of a variable load test unit. It can also be done in the form of suitably large power resistors. When using an oscilloscope to measure waveforms on the prototype board, the ground leads of the oscilloscope probes should be as short as possible and the area of the loop formed by the ground lead should be kept to a minimum. This will help reduce ground lead inductance and eliminate EMI noise that is not actually present in the circuit.



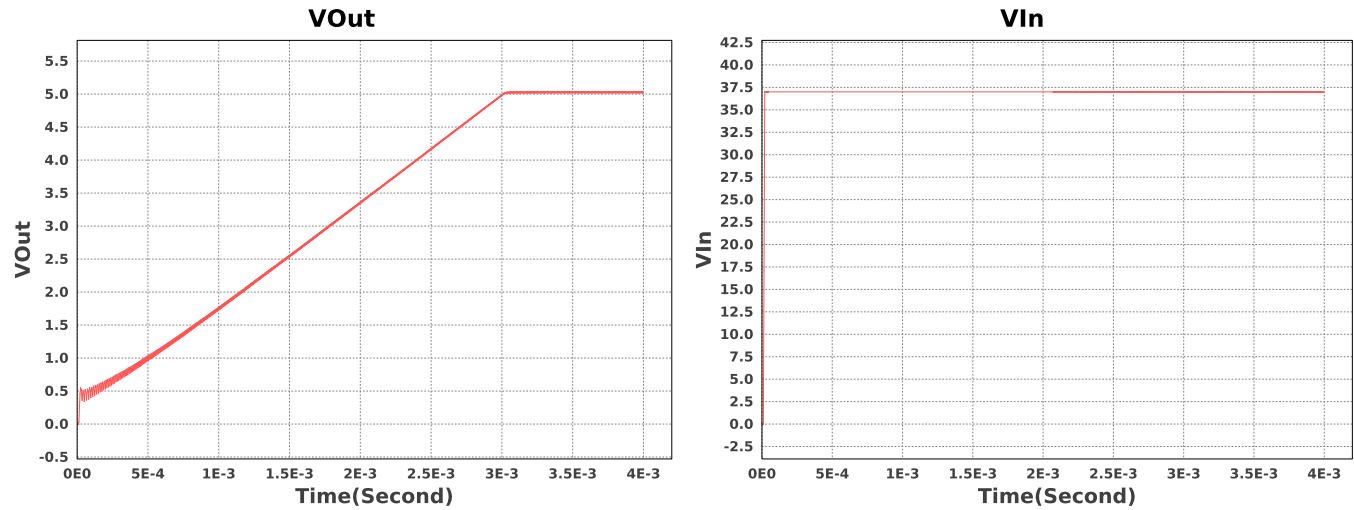
WEBENCH® Electrical Simulation Report

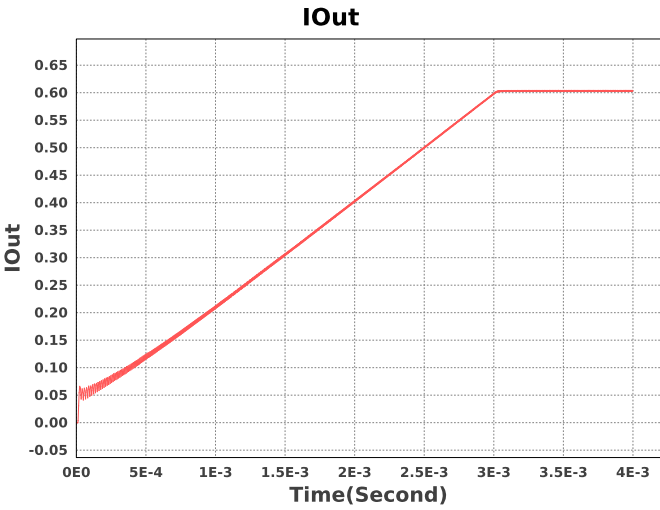
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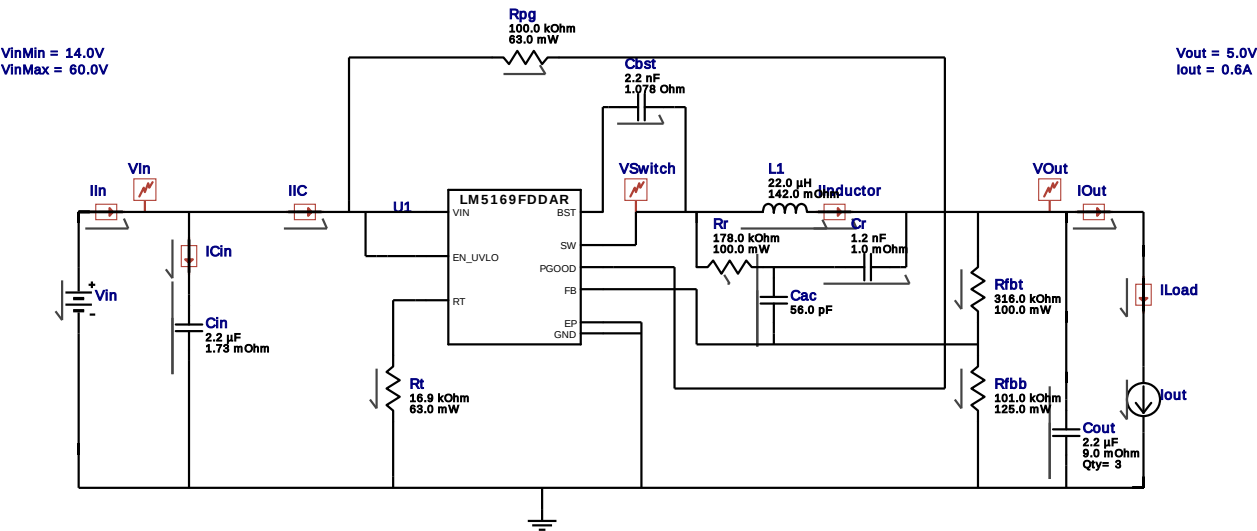
Simulation Parameters

#	Name	Parameter Name	Description	Values
1.	Rload	R	Load Resistance	8.333333333333334 Ohm



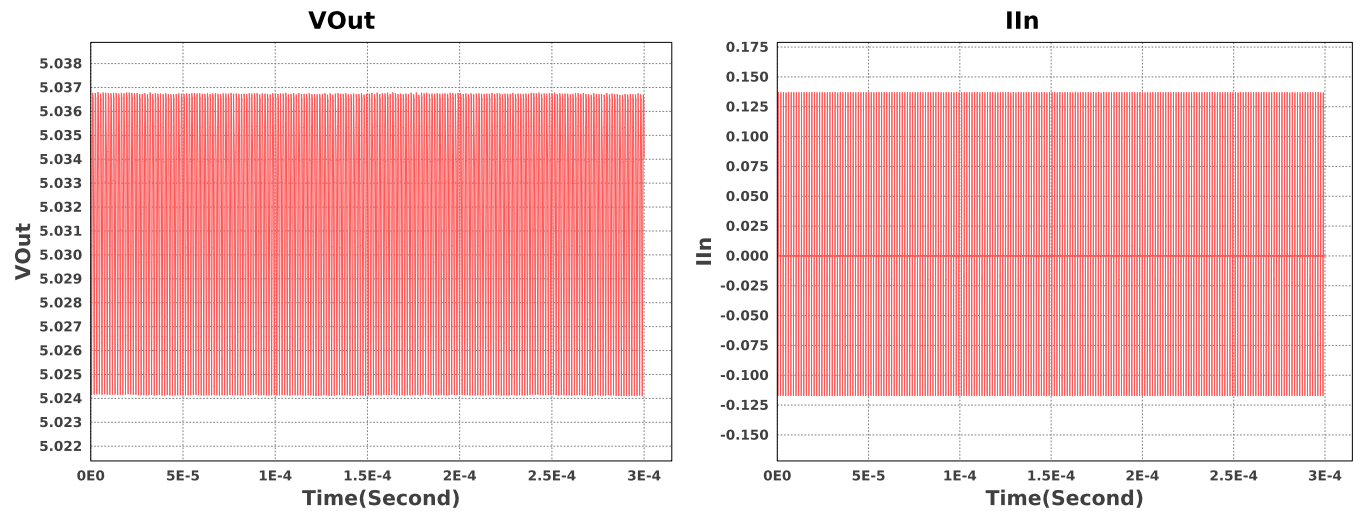


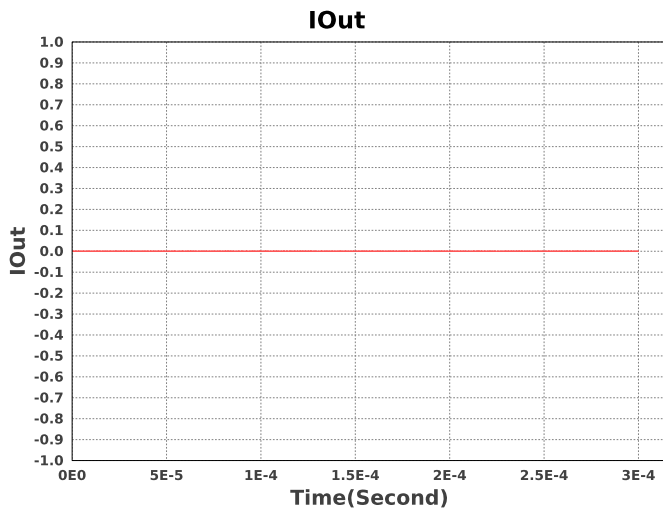
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Simulation Type = Steady State



Simulation Parameters

#	Name	Parameter Name	Description	Values
1.	Cbst	IC	Initial Condition	5
2.	Cout	IC	Initial voltage	Vout





Design Assistance

1. Master key : A621BC7ACCC4E41E9DAA23ECB35F628C[v1]
2. **LM5169F-DDA** Product Folder : <http://www.ti.com/product/lm5169> : contains the data sheet and other resources.

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